

Solution:

For the IBM corporation, the calculations are done as follows:

Issued shares = 2,182.5 million

Shares held in treasury = 1,019.3 million

Thus, the number of shares that are issued and outstanding are given by $2,182.5 - 1,019.3 = 1,163.2$ million.

Further, there are 4,687.5 million shares authorized.

Thus, the total number of unissued shares are given by $4,687.5 - 2,182.5 = 2,505$ million shares.